Friction: Test the Surfaces · Tier Pack · Aspire

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Aspire Route (Book First)

Explain why we repeat each surface and state one limitation of our two-surface comparison.

Help: Repeating checks consistency. Our results apply to this __ and release speed, not every possible surface.

Optional Make It · Aspire

Make an uncertainty audit card

Write a 3-point check card explaining why scientists repeat trials, keep outliers, and state limits on their conclusions.

Word Help

run-out surface: the flat test surface at the bottom of the ramp where the block slows and stops

stopping distance: how far the block travels along the run-out surface before stopping, in cm

control variable: a factor kept the same so the surface comparison is fair

repeat trial: running the same surface test more than once to check consistency

Exit Check

Which stopping distance shows more friction: 24 cm or 46 cm?

Name two things we kept the same to make the surface test fair.

Why should we keep a surprising result instead of rubbing it out?